

Stony Brook University
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ASSIGNMENT 3

SOLUTIONS

INSTRUCTIONS: Show all your work to get full points. Please also cut and paste at the end of your submission the R code you have used for the last two problems.

1. Simple Regression in a Spreadsheet. Suppose you have the following data on 11 student's combined SAT scores X and their cumulative grade point average at graduation Y .¹

Y_i	X_i
3.63	1490
2.37	1300
3.33	1510
3.32	1420
3.27	1490
2.37	1180
3.61	1550
3.23	1460
2.59	1300
3.30	1450
3.21	1550

- Write down the econometric model relating X and Y (i.e. the linear regression model).
- Calculate the following:

$$n; \bar{X}; \bar{Y}; \sum (X_i - \bar{X})(Y_i - \bar{Y}); \sum (X_i - \bar{X})^2; \hat{\beta}_0 \text{ and } \hat{\beta}_1.$$

- Calculate $\hat{Y}_i$ and $\hat{u}_i$ for each observation.
- Calculate R^2 .
- Compute $\sum \hat{u}_i$. Explain why do you get this result.

¹Do not do this exercise by hand. Please do this in any Spreadsheet software that you are comfortable with. Another option would be to do it R. In any case you **should not use** any predefined command to calculate the OLS intercept and slopes.

Solution

(a) The linear regression model relating X and Y is

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

(b) It is useful to construct a table:

	Y_i	X_i	$(X_i - \bar{X})$	$(Y_i - \bar{Y})$	$(X_i - \bar{X})^2$	$(X_i - \bar{X})(Y_i - \bar{Y})$
1	3.62	1490	63	0.51	3935	31.9
2	2.37	1300	-127	-0.74	16198	94.3
3	3.33	1510	83	0.22	6844	18.1
4	3.32	1420	-7	0.21	53	-1.5
5	3.27	1490	63	0.16	3935	10.0
6	2.37	1180	-247	-0.74	61144	183.2
7	3.61	1550	123	0.50	15062	61.3
8	3.23	1460	33	0.12	1071	3.9
9	2.59	1300	-127	-0.52	16198	66.3
10	3.3	1450	23	0.19	517	4.3
11	3.21	1550	123	0.10	15062	12.2
MEAN	3.11	1427	0	0	12729	44
$\sum_{i=1}^{11}$	34.22	15700	0	0	140018	484

	$\hat{Y}_i$	$\hat{u}_i$	$(Y_i - \bar{Y})^2$	$\hat{Y}_i - \bar{Y}$	$(\hat{Y}_i - \bar{Y})^2$	$\hat{u}_i^2$
1	3.33	0.29	0.26	0.22	0.047	0.085
2	2.67	-0.30	0.55	-0.44	0.193	0.091
3	3.40	-0.07	0.05	0.29	0.082	0.004
4	3.09	0.23	0.04	-0.03	0.001	0.055
5	3.33	-0.06	0.03	0.22	0.047	0.003
6	2.26	0.11	0.55	-0.85	0.730	0.013
7	3.54	0.07	0.25	0.42	0.180	0.006
8	3.22	0.01	0.01	0.11	0.013	0.000
9	2.67	-0.08	0.27	-0.44	0.193	0.007
10	3.19	0.11	0.04	0.08	0.006	0.012
11	3.54	-0.33	0.01	0.42	0.180	0.106
MEAN	3	0	0.19	0	0.152	0.035
$\sum_{i=1}^{11}$	34	0	2.054	0	1.673	0.382

Then looking at the tables above we have:

- $n = 11$
- $\bar{X} = 1427$

- $\bar{Y} = 3.11$
- $\sum (X_i - \bar{X})(Y_i - \bar{Y}) = 484$
- $\sum (X_i - \bar{X})^2 = 140018$
- $\hat{\beta}_0 = -1.82$ and $\hat{\beta}_1 = 0.003$

(c) See table above.

(d) From the second table above we know

$$TSS = 2.054, \quad ESS = 1.673, \quad SSR = 0.382.$$

Therefore using the formula for R^2 we get

$$R^2 = \frac{ESS}{TSS} = 0.81,$$

alternatively,

$$R^2 = 1 - \frac{SSR}{TSS} = 0.81.$$

(e) $\sum_{i=1}^{11} \hat{u}_i = 0$ (See the second table above).

This is always true when we run an OLS regression. In other words, this is an implication of the use of OLS to compute estimators. Since OLS seeks to minimize the distance between the observations and the regression line, what OLS is actually doing is minimizing the "mistakes" that our estimated model produces so that on average this mistake is zero or equivalently:

$$\frac{\sum_{i=1}^n \hat{u}_i}{n} = \frac{0}{n}$$

A more formal argument is as follows:

$$\begin{aligned} \sum_{i=1}^n \hat{u}_i &= \sum_{i=1}^n (Y_i - \hat{Y}_i) \\ &= \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) \\ &= \sum_{i=1}^n (Y_i - \bar{Y} + \hat{\beta}_1 \bar{X} - \hat{\beta}_1 X_i) \\ &= \sum_{i=1}^n Y_i - \sum_{i=1}^n \bar{Y} + \sum_{i=1}^n \hat{\beta}_1 \bar{X} - \sum_{i=1}^n \hat{\beta}_1 X_i \\ &= n\bar{Y} - n\bar{Y} + n\hat{\beta}_1 \bar{X} - n\hat{\beta}_1 \bar{X} \\ &= 0 \end{aligned}$$

Where we have used:

$$\frac{1}{n} \sum_{i=1}^n X_i = \bar{X} \Rightarrow \sum_{i=1}^n X_i = n\bar{X}$$

$$\frac{1}{n} \sum_{i=1}^n Y_i = \bar{Y} \Rightarrow \sum_{i=1}^n Y_i = n\bar{Y}$$

$$\sum_{i=1}^n c = nc,$$

for any constant c .

2. Consider the savings function

$$sav = \beta_0 + \beta_1 inc + u, \quad u = \sqrt{inc} \cdot e,$$

where e is a random variable $E(e) = 0$ and $\text{Var}(e) = \sigma_e^2$. Assume that e is independent of inc .

- Show that $E(u|inc) = 0$, so that the key *zero conditional mean assumption* is satisfied. *Hint: If e is independent of inc , then $E(e|inc) = E(e)$.*
- Show that $\text{Var}(u|inc) = \sigma_e^2 \cdot inc$. *Hint: $\text{Var}(e|inc) = \text{Var}(e)$ if e and inc are independent.*
- Provide a discussion that supports the assumption that the variance of savings increases with family income.

Solution

- $E(u|inc) = E(\sqrt{inc} \cdot e|inc) = \sqrt{inc}E(e|inc) = \sqrt{inc}E(e) = 0$ where the next to last equality is due to independence of e and inc and the last equality is by the assumption $E(e) = 0$.
- $\text{Var}(u|inc) = \text{Var}(\sqrt{inc} \cdot e|inc) = (\sqrt{inc})^2 \text{Var}(e|inc) = inc \cdot \text{Var}(e) = inc \cdot \sigma_e^2$, where the second equality is by properties of the variance, the third equality is by independence of e and inc and the last equality is by assumption.
- A large share of the spending of low income families is spent on *economic needs*: food, housing, clothing and other basic necessities. On the other hand, higher income families have more discretion about their spending, i.e. some families might choose to consume more while others choose to save more. Thus, this suggests that the variability of savings with respect to income increases with the level of income.

3. First Look at Regression Output. Suppose that a researcher, using data on class size (CS) and average test scores from 100 third-grade classes, estimates the OLS Regression

$$\widehat{TestScore} = 520.4 - \frac{5.82}{(20.4)} CS, \quad R^2 = 0.08, \quad SER = 11.5,$$

where the standard errors for the coefficient estimates are given by the numbers in parentheses.

- Suppose a classroom has 20 students. What is the regression's prediction for that classroom's average test score?
- Last year a classroom had 19 students, and this year it has 21 students. What is the regression's prediction for the change in the classroom average test score?
- The sample average class size across the 100 classrooms is 21.4. What is the sample average of the test scores across the 100 classrooms? (*Hint: Review the formulas for the OLS estimators*).
- What is the sample standard deviation of the test scores across the 100 classrooms? (*Hint: Review the formulas for the R^2 and SER*).

- (e) Construct a 95% confidence interval for β_1 , the regression slope coefficient.
- (f) Calculate the p -value for the two-sided test of the null hypothesis $H_0 : \beta_1 = 0$. Do you reject the null hypothesis at the 5% level? At the 1% level?
- (g) Calculate the p -value for the two-sided test of the null hypothesis $H_0 : \beta_1 = -5.6$. Without doing any additional calculations, determine whether -5.6 is contained in the 95% confidence interval for β_1 .
- (h) Construct a 99% confidence interval for β_0 .

Solution

- (a) The regression's prediction for that classroom test score is

$$\widehat{TestScores} = 520.4 - 5.82 \times (20) = 404.$$

- (b) The prediction for last year was

$$\widehat{TestScores} = 520.4 - 5.82 \times (19) = 409.82.$$

while the prediction for this year is

$$\widehat{TestScores} = 520.4 - 5.82 \times (21) = 398.18.$$

Thus, the predicted change in the average test score in the classroom is

$$\Delta \widehat{TestScores} = 398.18 - 409.82 = -11.64.$$

Alternatively, using $\Delta \widehat{TestScores} = \hat{\beta}_1 \times \Delta \widehat{CS}$ we get

$$\Delta \widehat{TestScores} = -5.82 \times 2 = -11.64.$$

- (c) From the formula for $\hat{\beta}_0$ we have

$$\widehat{TestScores} = \hat{\beta}_0 - \hat{\beta}_1 \times \overline{CS} = 520.4 - 5.82 \times 21.4 = 395.852.$$

- (d) We can use the formula for the SER to calculate the sum of squared residuals SSR

$$SSR = (n - 1)SER^2 = 98 \times 11.5^2 = 12960.5$$

We can use the formula for R^2 to compute the total sum of squares TSS

$$TSS = \frac{SSR}{1 - R^2} = \frac{12960.5}{1 - 0.08} = \frac{12960.5}{0.92} = 14086.95.$$

Thus, the sample variance and standard deviation are

$$s^2 = \frac{TSS}{n - 1} = \frac{14086.95}{99} = 142.29, \quad s = \sqrt{142.29} = 11.92.$$

- (e) $[-5.82 \pm 1.96(2.21)] = [-10.152, -1.488].$

(f) The t -statistic is

$$t = \frac{-5.82}{2.21} = -2.633,$$

and the p -value is $2\Phi(-2.633) = 0.008$. Since the p -value is less than 5% and 1%, we reject the null hypothesis at both levels.

(g) The t -statistic is

$$t = \frac{-5.82 - (-5.6)}{2.21} = -0.0995,$$

and the p -value is $2\Phi(-0.0995) = 0.9207$. Since we would not reject at the 5% level since the p -value is higher, then we know that -5.6 is contained in the 95% confidence interval. In fact, notice that this is indeed the case, we calculated the 95% CI to be: $[-10.152, -1.488]$.

(h) $[520.4 \pm 2.58(20.4)] = [467.768, 573.032]$.

4. Suppose that a researcher, using wage data on 250 randomly selected male workers and 280 female workers, estimates the OLS regression

$$\widehat{Wage} = 12.52 + \frac{2.12}{(0.23)} Male, \quad R^2 = 0.06, \quad SER = 4.2,$$

where $Wage$ is measured in dollars per hour and $Male$ is a binary variable that is equal to 1 if the person is male and 0 if the person is a female. Define the wage gender gap as the difference in mean earnings between men and women.

- What is the estimated gender gap?
- Is the estimated gender gap significantly different from 0? (Compute the p -value for testing the null hypothesis that there is no gender gap.)
- Construct a 95% confidence interval for the gender gap.
- In the sample, what is the mean wage of women? Of men?
- Another researcher uses these same data but regresses $Wages$ on $Female$, a variable that is equal to 1 if the person is female and 0 if the person is a male. What are the regression estimates calculated from this regression?

Solution

- Recall that for a regression equation of the form $Wage = \beta_0 + \beta_1 Male + u$, the gender gap is defined as $E(Wage|Male) - E(Wage|Female) = \beta_1$. Thus, the estimated gender gap is $\hat{\beta}_1 = 2.12$.
- The null hypothesis is $H_0 : \beta_1 = 0$ vs the alternative $H_a : \beta_1 \neq 0$. The t -statistic is

$$t = \frac{2.12}{0.36} = 5.89,$$

the p -value is $2\Phi(-5.89) = 3.86 \times 10^{-9}$. This p -value is less than 0.01, so we reject the null hypothesis that there is no wage gender gap at the 1% level.

- $[2.12 \pm 1.96(0.36)] = [1.41, 2.82]$.
- The sample average wage of women is $\hat{\beta}_0 = \$12.52/\text{hour}$. The sample average wage of men is $\hat{\beta}_0 + \hat{\beta}_1 = \$12.52 + \$2.12 = \$14.64/\text{hour}$.
- The regression model relating wages to gender can be written as either $Wage = \beta_0 + \beta_1 Male + u$ or as $Wage = \gamma_0 + \gamma_1 Female + v$. In the first regression equation, the variable $Male$ equals 1 for men and 0 for women; correspondingly, in the second regression $Female$ equals 1 for women and 0 for men.

Thus, in the first regression β_0 is the population mean of wages for women and $\beta_0 + \beta_1$ is the population mean wages for men. In the second regression equation, γ_0 is the population mean wages for men and $\gamma_0 + \gamma_1$ is the population mean wages for women. Therefore, the two set of coefficients in the two regression equations are related by the following equations:

$$\gamma_0 = \beta_0 + \beta_1.$$

$$\gamma_0 + \gamma_1 = \beta_0.$$

Given the estimated coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$, we have that

$$\hat{\gamma}_0 = \hat{\beta}_0 + \hat{\beta}_1 = 14.64,$$

$$\hat{\gamma}_1 = \hat{\beta}_0 - \hat{\gamma}_0 = -\hat{\beta}_1 = -2.12.$$

Given the relationship between the coefficient estimates, for each individual observation, the OLS residual is the same under the two regression equations, so that $\hat{u}_i = \hat{v}_i$. Thus the sum of squared residuals is the same under the two regressions. This implies that both the SER and the R^2 remain unchanged.

Summarizing, when regressing *Wages* on *Female*, we will get

$$\widehat{Wage} = 14.64 - 2.12Male, \quad R^2 = 0.06, SER = 4.2,$$

5. Empirical Exercise: Simple Linear Regression. To solve this problem you need to use the same dataset as in the second assignment (i.e., CPS08) posted on Blackboard. The dataset contains information on wages, gender, education level, and type of employment from 500 employees. The goal of this exercise is to explore the relationship between earnings and age.

Assume that the four OLS assumptions discussed in class are satisfied. Estimate the following simple regression model of *LogWage* on *Age*

$$\text{LogWage}_i = \beta_0 + \beta_1 \text{Age}_i + u_i, \quad i = 1, \dots, N.$$

- Report the estimates for $\hat{\beta}_0$ and $\hat{\beta}_1$. Interpret the estimate for the slope parameter.
- Compute the R^2 for this regression? How can you interpret its value?
- Repeat parts (a) and (b) but using the observations for female workers (i.e., $FEMALE = 1$).
- Repeat parts (a) and (b) but using the observations for male workers (i.e., $FEMALE = 0$).

6. Empirical Exercise. Use the data in COUNTYMURDERS to answer these questions. Use only the data for 1996. The variable *murders* is the number of murders reported in the county. The variable *execs* is the number of executions that took place of people sentences to death in a given county. Most states in the United States have the death penalty, but several do not.

- How many counties are there in the data set? How many counties had zero murders in 1996? How many counties had at least one execution? What is the largest number of executions? Why is the average number of executions so small?
- Estimate the equation

$$\text{murders} = \beta_0 + \beta_1 \text{execs} + u$$

by OLS and report the results in the usual way, including sample size and R^2 .

- Interpret the slope coefficient in part (b). Does the estimated equation suggest a deterrent effect of capital punishment?

- (d) What is the smallest number of murders that can be predicted by the equation? What is the residual for a county with zero executions and zero murders?
- (e) Explain why a simple regression analysis is not well suited for determining whether capital punishment has a deterrent effect on murders.